

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Master of Business Administration (MBA)
Semester –II University Examination May 2010

MB731: Costing and Control System

Date :18/05/2010, Tuesday

Time : 12:30 p.m. to 03:30 p.m.

Maximum Marks: 70

Instructions:

1. This paper contains six questions, of which the last one is a case.
2. All questions are compulsory.
3. Each of the first five questions will carry 10 marks and the last question will carry 20 marks.

Q-1 (A) Riddhi Ltd. provided you the following information: [05]

Production capacity	80%	60%
Costs (Rs. in lakhs)		
Direct Material	2.00	1.50
Direct Labour	2.00	1.50
Direct Expenses	1.60	1.20
Manufacturing expenses	4.00	3.85
Administrative and selling expenses	8.00	7.55
Sales	20.00	15.00

Find out fixed and variable portion of manufacturing, administrative and selling expenses. Semi-variable expenses are constant between 50% and 85% of capacity and increasing by 20% between 85% and 100% capacity. Prepare the flexible budget showing profit/loss at 50% and 90% capacity.

(B) A product passes through two processes. The output of process I becomes input of process II and output of process II is transferred to warehouse. The quantity of raw materials introduced into process I is 5,000 units at Rs. 2 per unit. The cost and output data for the month under review are as under: [05]

Particulars	Process I	Process II
Direct labour (Rs.)	3,000	5,000
Direct expenses (Rs.)	9,750	9,910
Production overhead (Rs.)	Rs. 16,000 recovered as per percentage of direct labour	
Normal loss (% of input)	5%	10%
Output in units	4,700 units	4,300 units
Loss realization (scrap)	Rs 1.00 per unit	Rs 5.00 per unit

There was no opening or closing work-in-progress stock.

Prepare Process I, Process II and abnormal loss/gain account if any.

Q-2 (A) In processing basic raw material, three joint products 'P', 'Q' and 'R' are produced after incurring joint cost of Rs. 510,000. All the three products are produced further after separation and sold as per detail given below: [02]

	PARTICULARS	P	Q	R	TOTAL
	Output (units)	10,000	20,000	30,000	
	Selling price /unit	Rs. 30	Rs. 20	Rs. 10	
	(a) SALES (Rs.)	?	?	?	?
	(b) Estimated profit (Rs.)	30,000	80,000	90,000	200,000
(a-b)	(c) COST OF SALES (Rs.)	270,000	320,000	210,000	800,000
	(d) Selling expense (Rs.)	?	?	?	?
(c-d)	(e) TOTAL COST(COGS)	?	?	?	?
	Further processing cost per unit	Rs. 7	Rs. 6	Rs. 2	
	(f) Further processing cost (Rs.)	?	?	?	250,000
(e-f)	(g) JOINT COST	?	?	?	510,000

Assume that selling expenses are apportioned over the product as a percentage of cost of sales. You are required to fill up the question mark places by preparing a statement showing apportionment of joint cost.

(B) A company currently operating at 80% capacity has the following particulars: [08]

Sales	Rs 6,400,000
Direct Material	Rs. 2,000,000
Direct labour	Rs. 800,000
Variable O/h	Rs. 400,000
Fixed O/h	Rs. 2,600,000

An export order has been received that would utilize half of the capacity of the factory. The order cannot be split, i.e. it has either to be taken in full and executed at 10% below the normal domestic prices or rejected totally.

The alternatives available to the management are:

1. Reject the order and continue with the domestic sales only (as at present), or
2. Accept the order, split capacity between overseas and domestic sales and turn away excess domestic demand, or
3. Increase capacity so as to accept the export order and maintain the present domestic sales by:
 - Buying equipment that will increase capacity by 10%. This will result in an increase of Rs. 200,000 in fixed cost, and
 - Work overtime to meet balance of required capacity. In that case, labour will be paid at one and a half times the normal wage rate.

Prepare a comparative statement of profitability and suggest the best alternative.

Q-3 (A) From the following records you are required to compute the [05]

- (1) Material variances
- (2) Labour variances

Particulars	Standard	Actual
Material:		
Input	7,000 kg.	10,000 kg.
Price per kg.	Rs. 20	Rs. 21
Output (units)	700,000	900,000
Labour:		
wage rate per employee per day	Rs. 6	Rs. 6.5

Number of employees is 200. Standard daily output per employee is 100 units. Total number of days worked is 50. Idle time paid for and included above is $\frac{1}{2}$ day.

(B)

I) Rupa Manufacturing Company supplies you the following information:

[02]

Normal capacity of plant	10,000 units	Marginal cost per unit	Rs. 75
Fixed cost when plant is operating	Rs. 100,000	Estimated selling price per unit	Rs. 80
Fixed cost when plant is shut down	Rs. 80,000	Estimated sales volume at this selling price	5,000 units

Should the company continue production or should it be closed down? Why?

(II) Column X below presents certain cost items, whereas column Y shows the cost category/ related entity. The entries in the two columns are not matched. Match each of the cost items in column X with most closely related entity in column Y :

[03]

'X'	'Y'
(1) Advertisement & Research and development cost:- Cost category	(a) Floor area occupied
(2) Direct labour + direct expenses + production overhead	(b) Cost of asset value
(3) Provident fund contribution :-Basis for apportionment of overhead	(c) Discretionary cost
(4) Repair and maintenance of factory building :- Basis for apportionment of o/h	(d) Conversion cost
(5) Investment centre performance measured by	(e) No. of workers
(6) Wages paid to the workers :-Cost category	(f) Wages of worker
	(g) ROI, EVA
	(h) Out of pocket cost

Q-4

(A) Following are the details related to a company which manufactures and sells only two products- Minor and Major.

[05]

Particular	Minor	Major
Selling price (per unit)	Rs. 10	Rs. 120
Variable cost (per unit)	Rs. 2	Rs. 100
Fixed cost (per unit)	Rs. 3	Rs. 10
Total capacity	10,000 units	4,000 units
Total capacity in use	60%	80%

The production capacity is interchangeable across the products; the company rewards its sales people purely on certain fixed percentage on

sales. While evaluating the latest year's financial performance, the company found that the sales have increased, but profits have decreased.

You are required to:

1. Identify breakeven point in units and rupees, P/V ratio, for both products.
2. Identify the reason for sales profit anomalic relation.
3. Suggest an appropriate measure to correct the situation.

(B) Explain the concept of Life cycle costing.

[03]

(C) Answer the following questions:

[02]

- (1) "Fixed cost is variable and variable cost is fixed." Explain the circumstances with example.
- (2) If selling price per unit after further processing is Rs. 12, further processing cost is Rs.6 per unit and selling price per unit at split off point is Rs. 8, should company sell its product at split off point or should it be further processed? Help the company to make decision with justification.

Q-5

T.K. Stores wants information about the profitability of individual product lines i.e. Soft drinks, Fresh Produce and Packaged Food. T.K. Stores has the following data for the year 2007-08 for each product line:

[10]

Particulars	Soft drinks	Fresh Produce	Packaged Food	Total
Revenues	Rs. 793,000	Rs. 2,100,600	Rs. 1,209,900	Rs. 4,103,500
Cost of goods sold	Rs. 600,000	Rs. 1,500,000	Rs. 900,000	Rs. 3,000,000

The stores also provide the following information of support cost (all cost other than cost of goods sold) for the 2007-08.

Activity	Description of activity	Total cost (Rs.)	Cost- allocation base
Bottles returns	Return of empty bottles	12,000	Directly tracing to soft drink line only
Ordering	Placing of order for purchase	156,000	1,560 purchase order
Delivery	Physical delivery and receipt of goods	252,000	3,150 deliveries
Self stocking	Stocking of goods on store shelves and on-going restocking	172,800	8,640 hours of self stocking
Customer support	Assistance provided to customers including checkouts	307,200	1,536,000 items sold
Total support cost		900,000	

Details about activity consumed by different stores are as follows:

Particulars	Soft drinks	Fresh Produce	Packaged Food	Total
Cost of bottle returned	Rs. 12,000	-	-	Rs. 12,000
Number of purchase orders placed	360	840	360	1,560
Number of deliveries received	300	2,190	660	3,150
Hours of self stocking time	540	5,400	2,700	8,640
Item sold	126,000	1,104,000	306,000	1,536,000

Questions:

- T.K. Stores currently allocates support costs (all cost other than cost of goods sold) to product lines on the basis of cost of goods sold of each product line. Calculate operating income for each product line.
- If T.K. Stores allocates support cost to product lines using an activity based costing system, calculate the operating income for each product line.
- Compare and comment on your answers for (i) and (ii).

Q-6

Analyse the given case and answer the questions given at the end:

[20]

Nisarg Corporation is a valve component manufacturing company. Division A of Company is the only source of supply for an intermediate product that is converted by Division B into saleable final product. A substantial part of Division A's cost is fixed. For any output up to 1,000 units a day, their total costs are Rs. 5,000 a day. Total cost increases by Rs. 1,000 a day for every additional 1,000 units made. Division A judges that its own result will be optimized if it sets its price at Rs 4 per unit. Statement of profit at different level of output for Division A has been given below in Exhibit-1.

Division B incurs additional cost in converting the intermediate product supplied by A into the finished product. These costs are Rs. 12,500 for any output up to 1,000 units and Rs. 2,500 per thousand for output in excess of 1,000 units. On the revenue side, B can increase its revenue only by spending more on sales promotion and by reducing selling prices. Its sales forecast is shown in following table [Exhibit 2 of column (5)].

Based on the above details, statement of profit is set out for Division A in Exhibit 1 and Division B in Exhibit-2

Exhibit 1

Division A's output (units)	Division A's cost (production cost)	Selling price (Transfer price @ Rs. 4 per unit)	A's profit= Sales-Cost
1,000	5,000	4,000	(1,000)
2,000	6,000	8,000	2,000
3,000	7,000	12,000	5,000
4,000	8,000	16,000	8,000
5,000	9,000	20,000	11,000
6,000	10,000	24,000	14,000

Exhibit-2

Division B's output (units)	B's own process- ing cost (Rs.)	Raw material cost from A [@transfer price –Rs. 4 per unit]	B's total cost (Rs.)	B's revenue (sales forecast)- less selling expenses per 1000 units (Rs.)	B's total revenue less selling expenses (Rs.)	B's profit (Rs.)
(1)	(2)	(3)	(4)=(2)+(3)	(5)	(6)=(1) x (5)	(7)=(6)-(4)
1,000	12,500	4,000	16,500	17,500	17,500	1,000
2,000	15,000	8,000	23,000	13,250	26,500	3,500
3,000	17,500	12,000	29,500	11,000	33,000	3,500
4,000	20,000	16,000	36,000	9,250	37,000	1,000
5,000	22,500	20,000	42,500	8,000	40,000	(2,500)
6,000	25,000	24,000	49,000	6,667	40,000	(9,000)

Exhibit 2 makes it clear that the most profitable policy for Division B, in the circumstances, is to set its output at 2,000 or 3,000 units a day and to accept a profit of Rs.3500 a day. If its output is more than 3,000 or less than 2,000 it will make even less profit.

With division B taking 3,000 units a day from it, Division A's revenue, at Rs. 4 per unit is Rs. 12,000 and its costs are Rs. 7,000. Hence A's profit is Rs 5,000 a day. Adding it to B's profit of Rs. 3,500, a day's aggregate profit for the corporation is Rs. 8,500 a day.

Answer the following questions:

1. Suppose, division B decides to sell 2000 units instead of 3,000 units which gives the same profit of Rs. 3,500, what would be the impact of this division A's profit. And how it will affect the profit of company?

Situation -1: Abandons its divisionalised structure

2. The instructions to the general managers of both divisions were initially to maximize the divisional profit for separate units under decentralised structure. Suppose, the company now abandons its divisionalised structure, and instead of having two profit centres, A and B, it combines them into a single profit centre, with responsibility for both production of intermediate and processing it to completion.

All other conditions previously present continue to apply. Thus the market conditions that formerly faced by Division B now confronts the single profit centre. Its cost is equal to combined cost of A and B, eliminating charges previously made by A to B for the supply of intermediate.

- (i) Prepare the schedule of cost and revenue for the single profit centre based on details given at exhibit 1 and 2.
- (ii) Find out the level of output at which this single profit centre (combination of Division A and B) earns the highest profit along with its profit.
- (iii) What is the difference of earning in amount under the changed situation (centralized single profit centre) compared to decentralised

structure (separate profit centre). What could be the reason for the difference?

- (iv) If Division A sells 4,000 units to division B, what should be optimum transfer price be?

Situation-2: Divisionalised structure

3. Assume divisionalised structure to maximize the division profit for separate units. Division A has also other product line for manufacturing premium grade valve component used for special purpose valves manufacturing. Assume also that Division A is not the only source of supply for the intermediate product to Division B.

Similarly Division 'B' manufactures special valves, in addition to general purpose valves. It can procure the components for special purpose valves from Division A or from elsewhere. The Division B has always acquired a 2,000 premium grade valve component from Division A at Rs. 300 per unit. However, when informed that the Division A was increasing its unit price to Rs. 325 per unit, the Division B management decided to purchase the component from outside suppliers at a price of Rs. 300.

The Division A had recently acquired some specialized equipment that was used primarily to make this component. The manager cited the resulting high depreciation charges as justification for price boost. He asked the president of company to instruct the Division B to buy from Division A at Rs. 325 price per unit. He supplied the following data to back his request:

Division B annual purchases of component	2,000 units
A's variable costs per unit	Rs. 285
A's fixed costs per unit	Rs. 30

You are required to answer following questions:

- Suppose there are no alternative uses of division A's facilities. Will the company as a whole benefit if the Division B buys from the outside suppliers at Rs. 300 per unit? Show computation to support your answer.
- Suppose internal facilities of division A would not otherwise be idle. The equipment and other facilities would be assigned to other production operations that would otherwise require an additional annual outlay of Rs. 40,500. Should the Division B purchase from outsiders at Rs. 300 per unit?
- Suppose that there are no alternative uses for Division A's internal facilities and that the outsider's selling price drops by Rs. 30. Should the Division B purchase from outsiders?
- As the president, how you respond to Division A manager's request? Would your response differ depending on the specific situations described in number (1) through (3) above? Why?

*** BEST OF LUCK ***